

Code-Layout, Readability and Reusability

Date	11 July 2026
Team ID	XXXXXX
Project Name	StudyAI – Smart AI Study Companion & Quiz Generator
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained throughout the code.	Yes	Code follows consistent formatting and indentation standards.
2	Proper File Structure	Project files and folders are logically organized into frontend and backend modules.	Yes	Uses a modular directory structure with React and Flask.
3	Meaningful Variable Names	Variables and constants are named according to their purpose.	Yes	Improves readability and maintainability.
4	Function / Method Names	Functions and methods have descriptive and meaningful names.	Yes	Follows standard naming conventions.
5	Code Comments	Important logic is documented using inline and block comments where necessary.	Yes	Key modules and complex logic are documented.
6	Modular Design	Code is divided into reusable components, services, routes, and utility modules.	Yes	Frontend components and backend services are modular.
7	No Redundant Code	Duplicate and unused code has been minimized or removed.	Yes	Code has been refactored to avoid redundancy.
8	Error Handling	Application handles exceptions and invalid inputs gracefully.	Yes	Input validation, API error handling, and exception handling are implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Python Flask, React.js	User Login, Registration, Protected Routes, JWT Authentication	High
2	API Client (Axios)	JavaScript (Axios)	All Frontend API Requests, Token Management, Error Handling	High
3	File Upload Module	React.js, Python Flask	Study Material Upload (PDF, DOCX, TXT), Text Processing	High
4	AI Service Module	Python Flask, Groq SDK	Summary Generation, Flashcards, Quiz Generation, Study Plans	High
5	Dashboard Components	React.js, Chart.js	Analytics Dashboard, Performance Tracking, Progress Visualization	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend, backend, services, routes, and components.
Readability	5	Code uses meaningful variable names, consistent formatting, and follows standard coding practices.
Reusability	5	Reusable React components, Flask services, API client, and utility modules reduce code duplication.
Documentation / Comments	4	Code includes comments for important modules and functions; additional documentation can be